Myopia 24, Autosomal Dominant (MYP24): A Comprehensive Disease Characterization

MONDO:0014411 · OMIM #615420 · Causal gene: *SLC39A5* (ZIP5) · HGNC:20502 · 12q13.3

Summary

Myopia 24, autosomal dominant (MYP24) is a rare Mendelian form of non-syndromic high (axial) myopia caused by heterozygous, damaging variants in ***SLC39A5***, the gene encoding the basolateral zinc-uptake transporter **ZIP5**. The disease was defined molecularly by Guo et al. in 2014, who used whole-genome linkage plus exome sequencing to identify a co-segregating nonsense mutation (c.141C>G, p.Y47*) in a multigenerational Chinese family with severe high myopia, with additional variants in independent families and sporadic cases [PMID: 24891338]. The disease name "Myopia 24" is a serial locus designation and does not itself name the gene; the correct and intended gene for MONDO:0014411 / OMIM #615420 is *SLC39A5*, and this report follows that identifier anchor throughout.

Mechanistically, MYP24 is a disorder of **scleral extracellular-matrix (ECM) homeostasis**. Reduced ZIP5 function lowers intracellular zinc in scleral fibroblasts, which destabilizes Smad proteins and inhibits BMP/TGF-β signalling, thereby decreasing transcription of collagen and other ECM genes (COL1A1, COL2A1, COL4A1, FN1, LAMA1). The weakened sclera permits excessive posterior-globe elongation, producing early-onset, progressive axial high myopia (spherical equivalent ≤ −6.00 D; axial length typically > 26 mm). Because axial elongation itself is the proximate driver of sight-threatening complications — myopic maculopathy, posterior staphyloma, choroidal neovascularization, retinal detachment, cataract, and glaucoma — the long-term morbidity of MYP24 is visual rather than life-limiting.

There is **no gene-specific or zinc-targeted therapy** for MYP24. Management mirrors general high-myopia care: optical refractive correction, pediatric axial-elongation control (low-dose atropine, orthokeratology, defocus/aspherical-lenslet spectacles), encouragement of protective outdoor time, and lifelong retinal surveillance for complications. The gene–disease

relationship, while reproducible, rests on a limited number of families and cohorts that are overwhelmingly East Asian (Chinese), and several reported variants remain candidate/VUS rather than definitively curated in ClinVar/ClinGen.

Key Findings

Finding 1 — MYP24 is caused by heterozygous *SLC39A5* (ZIP5) mutations

MYP24 (MONDO:0014411; OMIM #615420) is an autosomal dominant, non-syndromic high myopia caused by heterozygous mutations in ***SLC39A5*** on chromosome **12q13.3**, which encodes the zinc transporter **ZIP5**. The defining study combined whole-genome linkage analysis with exome sequencing in a non-syndromic severe high myopia family and identified a nonsense mutation co-segregating with the phenotype; the same variant appeared in an unrelated sporadic case, and a second missense variant co-segregated in a separate family. Neither variant was present in 1,276 controls.

> "We identified a nonsense mutation (c.141C>G:p.Y47*) in SLC39A5 co-segregating with the phenotype in a non-syndromic severe high myopia family. The same nonsense mutation (c.141C>G:p.Y47*) was detected in a sporadic case and a missense mutation (c.911T>C:p.M304T) was identified and co-segregated in another family" — PMID: 24891338

Consolidated gene identifiers: ***SLC39A5***, HGNC:20502, NCBI Gene 283375, Ensembl ENSG00000139053, UniProt Q6ZMH5, OMIM *608730; protein product ZIP5** (SLC39 / ZIP family zinc transporter).

Finding 2 — Disease mechanism: a zinc → Smad → TGF-β → scleral ECM axis

The causal variants disrupt a signalling cascade that controls scleral collagen/ECM synthesis. Wild-type (but not disease-associated) SLC39A5 regulates Smad1, a key phosphoprotein downstream of the BMP/TGF-β pathway, and SLC39A5 is abundantly expressed in the sclera and retina during eye development [PMID: 24891338]. Functional dissection in cells showed that SLC39A5 loss produces zinc deficiency that destabilizes Smad proteins, inhibiting TGF-β signalling and downstream ECM synthesis; re-expression rescues the phenotype [PMID: 34302427].

> "the SLC39A5 depletion-induced zinc deficiency destabilized Smad proteins, which inhibited the TGF-β signalling and downstream ECM synthesis, thus contributing to the pathogenesis" — PMID: 34302427
>
> "wild-type, but not disease-associated SLC39A5 inhibited the expression of Smad1, a key phosphate protein in the downstream of the BMP/TGF-β (bone morphogenic protein/transforming growth factor-β) pathway" — PMID: 24891338

SLC39A5-knockout HEK293 cells show migration/morphogenesis defects and reduced transcription of ECM genes COL1A1, COL2A1, COL4A1, FN1, and LAMA1, via blocked R-Smad phosphorylation [PMID: 34302427].

Finding 3 — Pathogenic variant spectrum

Reported heterozygous *SLC39A5* variants in non-syndromic high myopia span nonsense, frameshift, and missense classes, all rare/absent in population databases (gnomAD, 1000G, ExAC; AF < 0.01) and predicted damaging by multiple in-silico tools (SIFT, PolyPhen-2, PROVEAN, etc.):

Variant (cDNA)	Protein	Class	Source	PMID
c.141C>G	p.Y47*	Nonsense	Guo 2014	24891338
c.911T>C	p.M304T	Missense	Guo 2014	24891338
c.1238G>C	p.G413A	Missense	Jiang 2014	25525168
c.860C>T	p.P287L	Missense	Feng 2017	28442722
c.956G>C	p.R319T	Missense	Feng 2017	28442722
c.726dupA	p.Arg243fs*140	Frameshift	Liu 2020	32215939
c.145G>A	p.E49K	Missense	Ye 2023	36964802
c.325G>T	p.G109W	Missense	Ye 2023	36964802
c.577G>A	—	Missense	Liu 2025	40920702

> "one novel frameshift mutation (c.726dupA, p.Arg243fs*140) in SLC39A5" — PMID: 32215939

> "three novel missense mutations, c.860C > T, p.Pro287Leu and c.956G > C, p.Arg319Thr in SLC39A5" — PMID: 28442722

The combination of a clear nonsense/frameshift (loss-of-function) spectrum and the functional evidence that wild-type ZIP5 is required for Smad/TGF-β regulation supports a **loss-of-function** mechanism acting dominantly (haploinsufficiency or dominant-negative of the signalling module).

Finding 4 — Clinical phenotype: early-onset axial high myopia with retinopathy risk

High myopia is defined as refractive error ≤ −6.00 D. SLC39A5-related cases are early-onset (presenting < 7 years in early-onset high myopia [eo-HM] cohorts), with eo-HM probands showing median spherical equivalent roughly −9.5 to −11.25 D and median axial length ~26–27 mm [PMID: 36964802]. In a 731-patient cross-sectional study, SLC39A5 was among genes "more relevant to eye elongation," and axial length was quantitatively tied to downstream morbidity:

> "1-mm increase in AL conferred 10.84% higher risk of pathologic retinopathy (Category ≥2) as well as 7.35% higher risk of low vision (best-corrected visual acuities <0.3)" — PMID: 31560770
>
> "SLC39A5, CCDC111, BSG, and P4HA2 were more relevant to eye elongation" — PMID: 31560770

Phenotype characteristics: onset childhood/early-onset; severity moderate-to-severe; progression progressive during growth years; the primary phenotype is axial high myopia with age-accumulating complication risk. **Quality-of-life impact** is driven by refractive dependence and, in those who progress to pathologic myopia, by irreversible central vision loss.

Suggested HPO terms: HP:0011003 (High myopia); HP:0012801 (Increased axial length of the globe); HP:0000545 (Myopia); HP:0007917 (Posterior staphyloma); HP:0000541 (Retinal detachment).

Finding 5 — Inheritance, penetrance, and population distribution

MYP24 is autosomal dominant (OMIM #615420), co-segregating across multigenerational families [PMID: 24891338]. Reported families/probands are overwhelmingly East Asian (Chinese) cohorts. Monogenic high myopia genes collectively explain only a minority of eo-HM families, and SLC39A5 is one of >17 candidate genes:

> "Sixteen potentially pathogenic variants predicted to affect protein function in eight of seventeen causative genes for HM in fifteen (13.3%) families were revealed" — PMID: 36964802

High myopia overall is predominantly polygenic/multifactorial, placing MYP24 in context as a rare Mendelian subset:

> "The cross-ancestry meta-analysis identified 932 RE-associated variants" — PMID: 42009823

No gene-specific prevalence/incidence rate (per 100,000) exists; penetrance appears high within reported families but is not formally quantified; no founder effect, germline mosaicism, or consanguinity role has been specifically established for SLC39A5.

Finding 6 — ZIP5 is a basolateral zinc-uptake transporter controlling organismal zinc status

Human/mouse ZIP5 is a Zn(II)-specific uptake transporter of the SLC39/ZIP family. Unusually, it is **not** endocytosed/down-regulated by zinc repletion and localizes to the **basolateral** membrane of polarized cells, mediating serosal-to-mucosal zinc transport [PMID: 15322118]. It is expressed in intestine, pancreas, liver, and kidney (zinc-homeostatic tissues) and, importantly for disease, abundantly in sclera and retina during eye development [PMID: 24891338].

> "the endocytic removal of mZip5 from the plasma membrane is not triggered by zinc treatment. Thus, the activity of mZip5 does not appear to be down-regulated by zinc repletion" — PMID: 15322118

The ***Slc39a5*-knockout mouse** is viable and fertile but shows altered zinc homeostasis:

> "total knockout of ZIP5 led to increased zinc in the liver in mice fed a zinc-adequate (ZnA) diet but impaired accumulation of pancreatic zinc in mice fed a zinc-excess (ZnE) diet" — PMID: 24303081

Finding 7 — Scleral ECM/collagen weakening is the effector step

The sclera is the primary effector tissue. In myopia it undergoes ECM remodeling — changes in composition, collagen organization, and biomechanical integrity — producing axial elongation; experimental scleral-strengthening therapies (posterior scleral reinforcement, collagen crosslinking) are under investigation [PMID: 42309346]. SLC39A5/ZIP5 is abundantly expressed in the sclera and controls scleral-fibroblast ECM gene transcription via the zinc–Smad–TGF-β axis [PMID: 24891338; 34302427].

> "excessive axial elongation resulting in biomechanical weakening of the sclera, and increasing the risk of complications such as posterior staphyloma, myopic macular degeneration, retinal detachment, cataract, and glaucoma" — PMID: 42309346

Finding 8 — Complications and prognosis

Axial elongation biomechanically weakens the sclera, raising lifetime risk of posterior staphyloma, myopic macular degeneration, choroidal neovascularization, peripheral retinal degeneration, retinal detachment, cataract, and glaucoma [PMID: 42309346; 39036706]. Approximately one in three high-myopia patients develop visual impairment with age; anxiety is recognized as the first extra-ocular complication of high myopia [PMID: 39036706]. MYP24 is **not life-limiting**; its morbidity is visual.

> "one in three patients with high myopia develop visual impairment with age" — PMID: 39082137

Finding 9 — Environmental/gene–environment context and protective factors

Although MYP24 is Mendelian, the broader myopia phenotype is strongly modified by environment. Increased **outdoor time** is protective against rapid axial growth, plausibly via sunlight-driven retinal dopamine release; excessive near work and reduced outdoor exposure increase risk; parental myopia is a major risk factor/confounder, and urban living modestly raises risk.

> "actively engaging in outdoor activities had a protective effect in the 7-9-year group (OR, 0.65, 95% CI 0.43 to 0.97)" — PMID: 39890431
>
> "a compelling connection between increased outdoor time and a decreased risk of myopia in children. This protective effect may primarily be attributed to exposure to the characteristics of natural light (i.e., sunlight) and the release of retinal dopamine" — PMID: 38297353

There is **no infectious or toxic etiology**. In Mendelian MYP24, genetic liability is primary but environmental modifiers plausibly influence the rate of axial elongation.

Finding 10 — Management: no gene-specific therapy; standard myopia control

No approved gene-/zinc-targeted therapy exists. Management follows general high-myopia care: optical correction plus pediatric axial-elongation control with low-dose atropine (0.01–0.05%), orthokeratology, and defocus/aspherical-lenslet spectacles (DIMS, HAL). Combination therapy outperforms monotherapy:

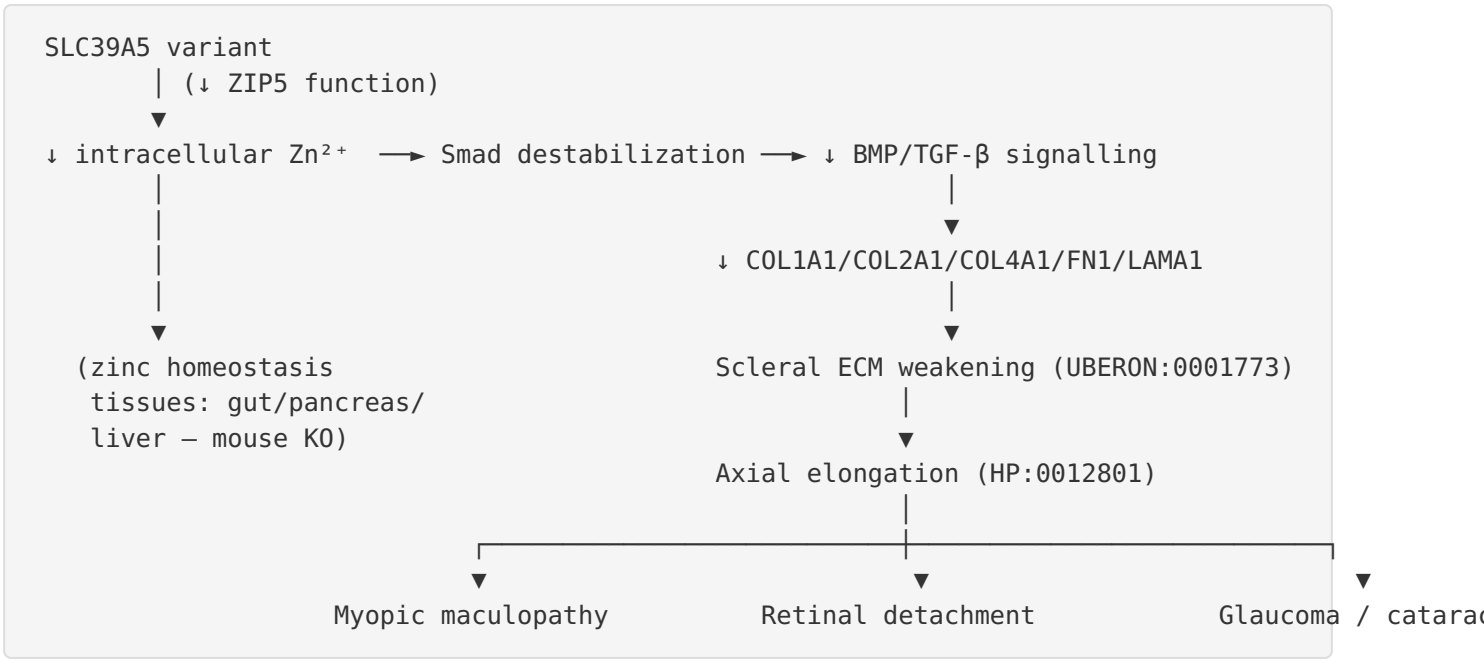
> "AL elongation was lower in the AOK group than in the OK group at 3, 6, and 12 months (0.03 ± 0.06 vs. 0.09 ± 0.04; 0.09 ± 0.08 vs. 0.18 ± 0.07; and 0.20 ± 0.12 vs. 0.33 ± 0.09 mm, respectively; all $P < 0.05$)" — PMID: 42770659

Pathologic complications require retinal surveillance and intervention. **Suggested NCIT terms:** C62043 (Atropine); treatment chemical entity CHEBI:16684 (atropine).

Mechanistic Model

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **heterozygous damaging *SLC39A5* variant** (nonsense/frameshift/damaging missense) **leads to** reduced functional ZIP5 zinc-transporter protein (loss of function acting dominantly).
2. Reduced ZIP5 activity in scleral fibroblasts **results in** decreased intracellular zinc availability (*inferred for sclera from cell models; demonstrated in vitro* [PMID: 34302427]).
3. Zinc deficiency **leads to** destabilization of Smad proteins [PMID: 34302427].
4. Smad destabilization **results in** impaired BMP/TGF-β signalling (blocked R-Smad phosphorylation; loss of wild-type SLC39A5's regulation of Smad1) [PMID: 24891338; 34302427].
5. Inhibited TGF-β signalling **leads to** reduced transcription of scleral ECM genes (COL1A1, COL2A1, COL4A1, FN1, LAMA1) [PMID: 34302427].
6. Reduced collagen/ECM synthesis **results in** scleral ECM remodeling and biomechanical weakening [PMID: 42309346].
7. Weakened sclera **leads to** excessive posterior-globe axial elongation (axial length > 26 mm; SE ≤ -6 D) [PMID: 31560770].
8. Axial elongation **results in** the clinical phenotype of early-onset high myopia and, cumulatively with age, **branches to** sight-threatening complications: myopic maculopathy, posterior staphyloma, choroidal neovascularization, retinal detachment, cataract, and glaucoma [PMID: 42309346; 39036706; 31560770].



Upstream vs downstream: the mutation and the zinc/Smad/TGF-β steps are upstream and molecular; scleral ECM weakening is the pivotal effector; axial elongation and retinal/choroidal complications are downstream and clinical.

Cell types / processes. Primary cell: scleral fibroblast (CL:0000057). Biological processes: GO:0005385 (zinc ion transmembrane transporter activity), GO:0007179 (transforming growth factor beta receptor signaling pathway), GO:0030198 (extracellular matrix organization). Chemical entities: CHEBI:29105 (zinc(2+)); CHEBI:16684 (atropine).

Anatomical and ontology crosswalk

Level	Structure	Ontology term
Organ (primary)	Posterior sclera	UBERON:0001773 (sclera)
Organ (secondary)	Retina	UBERON:0000966 (retina)
Tissue/cell	Scleral fibroblast	CL:0000057 (fibroblast)
Phenotype	High myopia	HP:0011003
Phenotype	Increased axial length	HP:0012801
Process	Zinc transport	GO:0005385
Process	TGF-β signalling	GO:0007179
Process	ECM organization	GO:0030198
Disease	MYP24	MONDO:0014411 / OMIM #615420
Treatment	Atropine	NCIT C62043 / CHEBI:16684

Identifier crosswalk

Identifier type	Value
MONDO	MONDO:0014411
OMIM (disease)	#615420 (MYP24)
Gene	SLC39A5
HGNC	HGNC:20502
NCBI Gene	283375
Ensembl	ENSG00000139053
UniProt	Q6ZMH5
OMIM (gene)	*608730
Locus	12q13.3
Protein	ZIP5
ICD-10 / ICD-11	H52.1 / 9D00.1
MeSH	D009216 (Myopia); D047475 (pathologic myopia)

Section-by-Section Notes

- **1. Disease information** — Non-syndromic AD axial high myopia; identifiers as tabulated above; information is derived from aggregated disease-level resources (OMIM, gene cohorts) plus individual-family genetic studies, not large EHR cohorts.
- **2. Etiology** — Primary cause is genetic (heterozygous SLC39A5 variants). Environmental modifiers (near work, outdoor time, parental myopia, urbanization) influence the broader myopia phenotype. No protective genetic variant is established; outdoor time is the best-supported protective environmental factor.
- **3. Phenotypes** — Dominated by high myopia (HP:0011003) and increased axial length (HP:0012801), childhood onset, progressive, with complication phenotypes developing with age.

- **4. Genetic/molecular** — See variant table; loss-of-function dominant mechanism; no established modifier genes, epigenetic changes, or chromosomal abnormalities specific to MYP24.
- **5. Environmental** — No toxic/infectious etiology; lifestyle factors modulate axial growth rate.
- **6. Mechanism** — See causal chain and diagram.
- **7. Anatomy** — Primary sclera (UBERON:0001773); secondary retina/choroid; scleral fibroblast (CL:0000057); bilateral involvement.
- **8. Temporal development** — Childhood/early onset; chronic, progressive during ocular growth; lifelong.
- **9. Inheritance/population** — AD; East-Asian–predominant reporting; rare Mendelian subset of high myopia.
- **10. Diagnostics** — Clinical: cycloplegic refraction, axial-length biometry, fundus/OCT for complications. Genetic: WES/targeted high-myopia gene panels including SLC39A5; single-gene/Sanger confirmation and segregation.
- **11. Outcome/prognosis** — Not life-limiting; visual morbidity accrues with axial elongation (~1 in 3 develop visual impairment).
- **12. Treatment** — Generic myopia control (atropine, ortho-K, DIMS/HAL); no gene-specific therapy; retinal intervention for complications.
- **13. Prevention** — Outdoor time; axial-growth control in children; genetic counseling (50% offspring risk, AD).
- **14. Other species** — Orthologous mouse *Slc39a5*; no naturally occurring ocular SLC39A5 disease reported in animals.
- **15. Model organisms** — *Slc39a5*-knockout mouse (zinc homeostasis); ocular/axial phenotype not reported — a key model gap.

Evidence Base

PMID	Role	Contribution
24891338	Landmark (human + in vitro)	Defines SLC39A5 as the MYP24 gene; nonsense/missense variants; BMP/TGF-β–Smad regulation; scleral/retinal expression
34302427	Mechanistic (in vitro)	Zinc–Smad–TGF-β–ECM axis; SLC39A5 depletion reduces COL/FN/LAMA transcription; rescue on re-expression
25525168	Variant cohort (human)	SLC39A5 missense c.1238G>C/p.G413A among eoHM genes
28442722	Variant cohort (human)	Novel SLC39A5 missense variants in 187 high-myopia patients
32215939	Variant cohort (human)	Frameshift c.726dupA LoF variant
36964802	Variant cohort (human)	Monogenic genes explain ~13.3% of eo-HM families; additional SLC39A5 variants
40920702	Variant cohort (human)	SLC39A5 c.577G>A in eo-HM; rarity in gnomAD/ExAC/1000G
31560770	Phenotype/ epidemiology (human)	SLC39A5 tied to eye elongation; axial length → retinopathy/ low-vision risk
15322118	Protein biology (in vitro)	ZIP5 basolateral localization; not zinc-down-regulated
24303081	Model organism (mouse)	Slc39a5-KO zinc-homeostasis phenotype; pancreatic protection
42309346	Mechanism/therapy review	Scleral ECM remodeling as biomechanical basis; scleral-strengthening approaches
39036706	Clinical review	High-myopia complication spectrum; anxiety as extra-ocular complication
39082137	Clinical review	~1 in 3 develop visual impairment; prevalence context
42770659	Interventional (human)	Atropine + ortho-K combination slows axial elongation
42768333		Ortho-K + atropine vs DIMS + atropine for AL control

PMID	Role	Contribution
	Interventional (human)	
39890431	Epidemiology (human)	Outdoor time protective (OR 0.65) against rapid axial growth
38297353	Review	Mechanism of outdoor-time protection (sunlight → retinal dopamine)
42009823	GWAS (human)	932 refractive-error variants; polygenic context for Mendelian MYP24

Evidence source types: the gene–disease relationship is supported by **human clinical/genetic** studies (family segregation + case cohorts) and **in vitro functional** work; the available **model organism** is the Slc39a5-knockout mouse, which models systemic zinc handling but has not been reported to recapitulate the ocular/axial myopia phenotype.

Limitations and Knowledge Gaps

- **Narrow evidence base / ancestry bias.** Essentially all reported MYP24 families and cohorts are East Asian (Chinese). Prevalence, incidence, penetrance estimates, and variant frequencies outside this population are unknown. No formal epidemiological rate (per 100,000) exists for the SLC39A5-specific subtype.
- **Gene–disease validity caveat.** Several reported variants are candidate/VUS and are not yet curated as definitively pathogenic in ClinVar/ClinGen. A formal ClinGen gene–disease validity classification was not confirmed in this investigation.
- **Mechanism partly inferred.** The zinc→Smad→TGF-β→ECM chain is demonstrated in cell models (HEK293, SLC39A5-KO), not directly in human scleral tissue of patients; the scleral step is inferred from expression data plus general myopia scleral biology.
- **No disease-specific animal model of the ocular phenotype.** The Slc39a5-KO mouse characterizes systemic zinc handling, not axial myopia; phenotype recapitulation for the eye is unestablished.
- **Penetrance/expressivity quantitatively uncharacterized.** AD inheritance is clear, but numeric penetrance, age-dependence, and intrafamilial variability are not well quantified for SLC39A5 specifically.

- **No gene-specific therapeutics or biomarkers.** There is no validated molecular biomarker, no zinc-/TGF-β-targeted intervention, and no pharmacogenomic data for MYP24.
- **Cross-species/natural disease data absent.** No naturally occurring SLC39A5 ocular disease in companion animals or wildlife was identified (OMIA/veterinary data not located for this subtype).

Proposed Follow-up Experiments / Actions

1. **Engineer an eye-relevant model.** Generate scleral-fibroblast–specific or knock-in (patient-variant) mouse/zebrafish models and quantify axial length, refraction, and scleral collagen architecture to test whether SLC39A5 loss produces axial elongation in vivo.
2. **Human scleral validation.** Where tissue is available (e.g., surgical/post-mortem highly myopic sclera), measure zinc content, Smad phosphorylation, and ECM-gene expression to confirm the in-vitro chain in patient tissue.
3. **ClinGen curation.** Submit the aggregated variant/segregation evidence for formal ClinGen gene–disease validity classification and deposit variants in ClinVar with ACMG/AMP evidence codes.
4. **Ancestry diversification.** Screen SLC39A5 in non–East-Asian high-myopia cohorts to estimate the gene's contribution and variant spectrum across populations and establish penetrance.
5. **Therapeutic hypothesis testing.** Evaluate whether local zinc supplementation or TGF-β-pathway modulation can restore scleral ECM synthesis in model systems — a mechanism-rational, disease-specific avenue distinct from generic myopia control.
6. **Genotype-stratified natural history.** Follow SLC39A5-positive patients longitudinally for axial-elongation rate and complication incidence versus polygenic high myopia, to determine whether monogenic cases warrant intensified surveillance.

Evidence types: human clinical/genetic (segregation + cohorts), in vitro functional, model organism (mouse), and computational (variant prediction, GWAS context). All mechanistic and clinical claims are anchored to the PMIDs cited above.